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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

REVIEW -0

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BACSE202 LAB COMPONENT PROJECT

B. Tech [CSE & CSE-With Specialisation]

AY:2026-27

APPENDIX 1
(A typical Specimen of a cover page)

TITLE OF PROJECT REPORT:

MODERN AIRLINE LOGISTICS & SUSTAINABILITY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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AY: 2026-27

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ABSTRACT

The **Modern Airline Logistics and Sustainability Management System** is a database-driven system designed to manage and integrate major airline logistics operations through a centralized relational database. Airline operations involve the continuous coordination of aircraft, flights, crew, passengers, specialized cargo, emissions and unexpected operational disruptions. Managing these activities through disconnected systems can lead to data inconsistency, delays, cargo-handling errors and difficulty in obtaining timely operational information.

The proposed system uses a relational database to maintain accurate and structured information regarding the airline fleet, flight schedules, crew assignments, passenger loyalty tiers, baggage limits, specialized cargo and flight-level sustainability records. It also incorporates cold-chain cargo monitoring for temperature-sensitive shipments, carbon-emission logging for sustainability analysis, and disruption management for recording delays caused by weather, technical issues, air-traffic control, crew-related problems and security events.

The database consists of nine interconnected entities with appropriate primary keys, foreign keys, NOT NULL, UNIQUE, CHECK and DEFAULT constraints. The schema is designed according to relational database principles and aims to reduce redundancy while preserving referential integrity.

SQL operations including DDL, DML, SELECT queries, single-row functions, aggregate functions, GROUP BY, HAVING, subqueries and JOIN operations are implemented to demonstrate how the database can support both routine airline operations and management-level decision making.

ACKNOWLEDGEMENT

I would like to thank Dr. BOOMINATHAN P (HOD) for reviewing the entire project and guiding us as the team leader, and also for helping me coordinate my project. After the completion of my project, I would like to take this chance to express my sincere gratitude to my project guide, Dr. KAVITHA G, who has guided me thoroughly throughout my project development. I would like to express my gratitude to all our friends and teachers, who helped and cooperated directly or indirectly in the accomplishment of this project. Heartfelt appreciation to those who helped me through cooperative acts and logical devices to make this project effective and user-friendly.

INTRODUCTION OF THE PROJECT

Airline logistics is a complex and highly coordinated operational environment in which aircraft, flights, crew members, passengers, cargo and airport resources must be managed efficiently. Every flight requires accurate scheduling, aircraft assignment, crew allocation, cargo handling and monitoring. A small error in any one of these areas can result in delays, increased operational costs and disruption of the overall transportation network.

Traditional or fragmented information systems often maintain operational data in separate modules. Such systems can make it difficult to obtain a unified view of the current status of flights and logistics operations. Problems such as flight delays, inappropriate crew assignment, cargo-handling issues and inefficient fuel usage may therefore become difficult to monitor and analyse.

The proposed **Modern Airline Logistics and Sustainability Management System** provides a centralized database for airline logistics operations. The system manages aircraft information, flight schedules, crew assignments, passenger information, loyalty-based baggage allowances, specialized cargo, carbon emissions and operational disruptions.

The system gives special attention to three modern airline challenges:

1. **Cold-Chain Cargo Management** – monitoring temperature-sensitive medical and perishable cargo.
2. **Sustainability Management** – recording fuel consumption and carbon dioxide emissions for each flight.
3. **Disruption Management** – recording and analysing delays caused by weather, technical problems, ATC, crew, security and other operational events.

BACKGROUND OF THE PROJECT WORK

Airline operations have traditionally depended on multiple independent systems to manage flight schedules, fleet details, passenger information, crew assignments and cargo handling. As airline operations expanded, the amount of data generated by each flight increased significantly. Separate systems and manual records can result in duplication, inconsistency and difficulty in obtaining a complete view of airline logistics.

Modern air transportation also faces challenges that go beyond basic passenger scheduling. Airlines transport temperature-sensitive pharmaceutical products, food, biological samples and other specialized cargo that require controlled environmental conditions during transportation. Therefore, cargo information must include handling requirements and temperature-monitoring information.

Environmental sustainability has also become an important part of airline operations. Fuel consumption and carbon emissions must be monitored so that organizations can analyse the environmental impact of individual flights and routes.

Another major challenge is operational disruption. Flights may be delayed because of weather conditions, technical faults, air-traffic restrictions, crew availability or security-related problems. Recording these events in a structured database allows management to identify frequent causes of delay and analyse operational performance.

The proposed database applies **Entity-Relationship modelling and relational normalization principles** to create a centralized airline logistics system. The design uses primary and foreign keys to maintain relationships among entities and uses constraints to prevent invalid data from entering the database.

HARDWARE AND SOFTWARE REQUIREMENTS

- **Hardware Requirements:**

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 / equivalent	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	20 GB free space	256 GB SSD or higher
Display	1366 × 768	1920 × 1080
Input	Keyboard and mouse	Keyboard and mouse
Network	Wi-Fi/Ethernet	Stable broadband/LAN

- **Software Requirements:**

Component	Specification
Operating System	Windows 10/11, Linux or macOS
DBMS	MySQL 8.0
Query Interface	MySQL Workbench / MySQL Command Line Client
Query Language	SQL
ER Diagram Tool	draw.io / Lucidchart
Documentation	Microsoft Word / Google Docs
Browser	Google Chrome / Microsoft Edge

E-R DIAGRAM

Entity	Primary Key (PK)	Attributes	Foreign Key (FK)
AIRCRAFT	Aircraft_ID	Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status	—
FLIGHT	Flight_ID	Route_Code, Dep_Time, Arr_Time, Flight_Status	Aircraft_ID → AIRCRAFT
CREW	Crew_ID	Crew_Name, Role, Base_Location	—
FLIGHT_CREW	Assignment_ID	Shift_Hours	Flight_ID → FLIGHT, Crew_ID → CREW
LOYALTY_TIER	Tier_ID	Tier_Name, Extra_Baggage_kg	—
PASSENGER	Passenger_ID	Passenger_Name, Passport_No, Baggage_kg	Tier_ID → LOYALTY_TIER
SPEC_CARGO	Cargo_ID	Cargo_Description, Weight_kg, Category, Req_Temp_C, Current_Temp_C, Tracking_Status	Flight_ID → FLIGHT
EMISSIONS	Log_ID	Fuel_Liters, CO2_Tons, Logged_At	Flight_ID → FLIGHT
DISRUPTION	Event_ID	Delay_Min, Reason_Code, Event_Date, Event_Status	Flight_ID → FLIGHT

DESCRIPTION ABOUT DESIGN OF THE DATABASE

The database is designed using relational database principles with emphasis on integrity, consistency and reduction of redundancy. The schema contains nine interconnected entities representing major airline logistics operations.

The **FLIGHT** entity acts as the central operational entity. Each flight is associated with one valid aircraft through the Aircraft_ID foreign key. This ensures that a flight cannot be scheduled without an aircraft existing in the database.

The relationship between **FLIGHT and CREW** is many-to-many because one flight requires multiple crew members and one crew member may participate in multiple flights. This relationship is resolved through the FLIGHT_CREW associative entity.

The **LOYALTY_TIER** and **PASSENGER** relationship allows the system to determine additional baggage allowances based on passenger membership level.

The **SPEC_CARGO** entity stores information about specialized cargo associated with flights. The design supports medical, perishable, hazardous and standard cargo and allows

temperature-sensitive shipments to be monitored using required and current temperature values.

The **EMISSIONS** entity stores fuel consumption and carbon dioxide emission information for flights. This enables the airline to analyse environmental impact at flight and route level.

The **DISRUPTION** entity stores operational interruptions, including delay duration, reason and status. This supports analysis of delay patterns and operational performance.

Constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK and DEFAULT have been applied to maintain data validity. The design follows normalization principles so that information is stored in logically separated relations.

Relationships / Cardinality Table

Parent Entity	Relationship	Child Entity	Cardinality	Meaning
AIRCRAFT	assigned to	FLIGHT	1 : M	One aircraft can operate many flights over time; each flight uses one aircraft.
FLIGHT	assigned through	FLIGHT_CREW	1 : M	One flight can have multiple crew assignments.
CREW	assigned through	FLIGHT_CREW	1 : M	One crew member can be assigned to multiple flights.
FLIGHT	carries	SPEC_CARGO	1 : M	One flight can transport multiple cargo shipments.
FLIGHT	generates	EMISSIONS	1 : M	A flight can have one or more emission logs.
FLIGHT	experiences	DISRUPTION	1 : M	One flight can experience multiple disruption events.
LOYALTY_TIER	assigned to	PASSENGER	1 : M	One loyalty tier can contain many passengers.
FLIGHT	has	CREW	M : N	A flight can have many crew members and a crew member can work on many flights; resolved through FLIGHT_CREW.

RELATIONAL SCHEMA

AIRCRAFT(
  Aircraft_ID PK,
  Model,
  Capacity_kg,
  Fuel_Efficiency,
  Aircraft_Status
)

FLIGHT(
  Flight_ID PK,
  Route_Code,
  Dep_Time,
  Arr_Time,
  Aircraft_ID FK → AIRCRAFT(Aircraft_ID),
  Flight_Status
)

CREW(
  Crew_ID PK,
  Crew_Name,
  Role,
  Base_Location
)

FLIGHT_CREW(
  Assignment_ID PK,
  Flight_ID FK → FLIGHT(Flight_ID),
  Crew_ID FK → CREW(Crew_ID),
  Shift_Hours
)

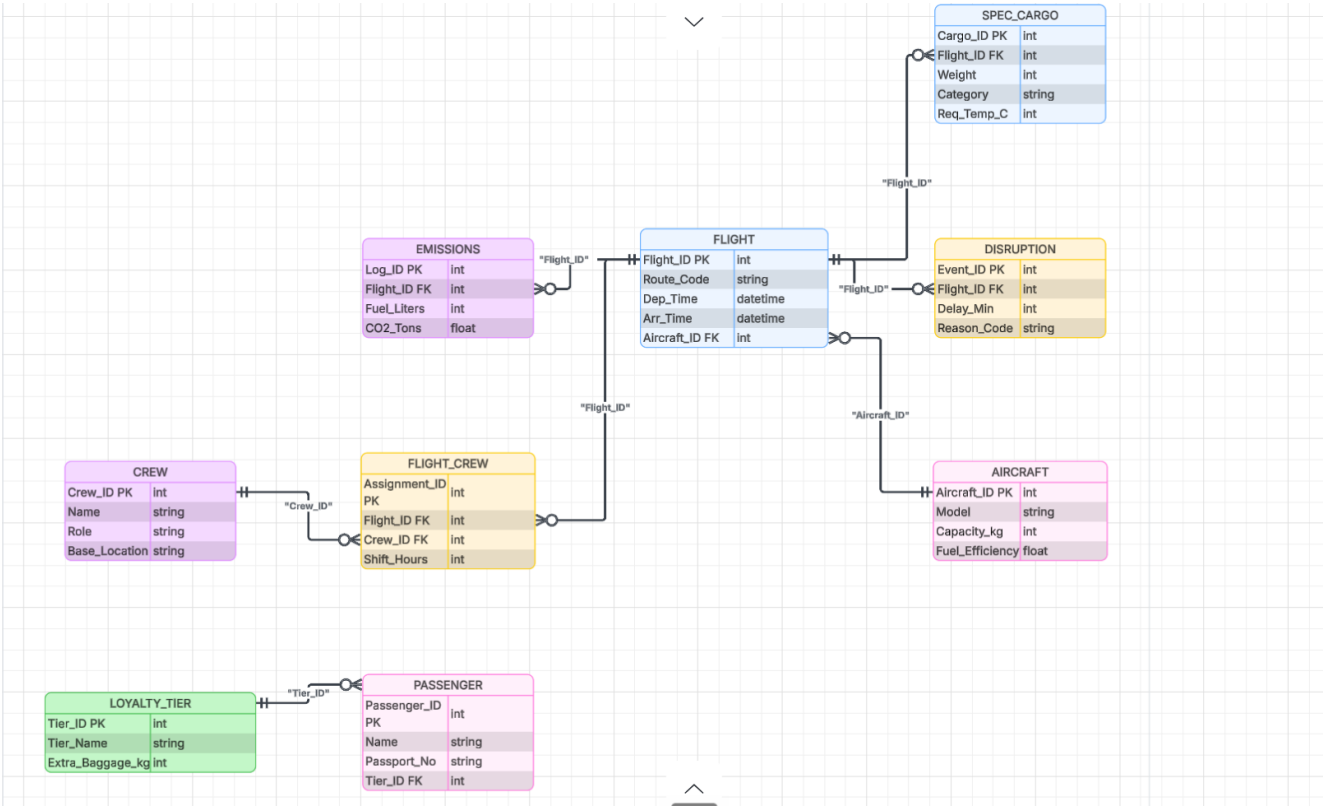
LOYALTY_TIER(
  Tier_ID PK,
  Tier_Name UNIQUE,
  Extra_Baggage_kg
)

PASSENGER(
Passenger_ID PK,
Passenger_Name,
Passport_No UNIQUE,
Tier_ID FK → LOYALTY_TIER(Tier_ID),
Baggage_kg
)

SPEC_CARGO(
Cargo_ID PK,
Flight_ID FK → FLIGHT(Flight_ID),
Cargo_Description,
Weight_kg,
Category,
Req_Temp_C,
Current_Temp_C,
Tracking_Status
)

EMISSIONS(
Log_ID PK,
Flight_ID FK → FLIGHT(Flight_ID),
Fuel_Liters,
CO2_Tons,
Logged_At
)

DISRUPTION(
Event_ID PK,
Flight_ID FK → FLIGHT(Flight_ID),
Delay_Min,
Reason_Code,
Event_Date,
Event_Status
)


SPEC_CARGO
Cargo_ID PK int
Flight_ID FK int
Weight int
Category string
Req_Temp_C int
"Flight_ID"
EMISSIONS
Log_ID PK int
Flight_ID FK int
Fuel_Liters int
CO2_Tons float
"Flight_ID"
FLIGHT
Flight_ID PK int
Route_Code string
Dep_Time datetime
Arr_Time datetime
Aircraft_ID FK int
"Flight_ID"
DISRUPTION
Event_ID PK int
Flight_ID FK int
Delay_Min int
Reason_Code string
"Flight_ID"
"Aircraft_ID"
CREW
Crew_ID PK int
Name string
Role string
Base_Location string
"Crew_ID"
FLIGHT_CREW
Assignment_ID PK int
Flight_ID FK int
Crew_ID FK int
Shift_Hours int
AIRCRAFT
Aircraft_ID PK int
Model string
Capacity_kg int
Fuel_Efficiency float
LOYALTY_TIER
Tier_ID PK int
Tier_Name string
Extra_Baggage_kg int
"Tier_ID"
PASSENGER
Passenger_ID PK int
Name string
Passport_No string
Tier_ID FK int

EXECUTION OF THE SQL QUERY

1. TABLE CREATION WITH ALL CONSTRAINTS

```
SQL> CREATE TABLE AIRCRAFT (
  2      Aircraft_ID NUMBER(10) PRIMARY KEY,
  3      Model VARCHAR2(50) NOT NULL,
  4      Capacity_kg NUMBER(10,2) NOT NULL,
  5      Fuel_Efficiency NUMBER(8,3) NOT NULL,
  6      Aircraft_Status VARCHAR2(20) DEFAULT 'ACTIVE' NOT NULL,
  7      CONSTRAINT chk_aircraft_capacity CHECK (Capacity_kg > 0),
  8      CONSTRAINT chk_aircraft_efficiency CHECK (Fuel_Efficiency > 0),
  9      CONSTRAINT chk_aircraft_status
 10          CHECK (Aircraft_Status IN ('ACTIVE', 'MAINTENANCE', 'RETIRED'))
 11  );

Table created.
```

```
SQL> CREATE TABLE LOYALTY_TIER (
  2      Tier_ID NUMBER(10) PRIMARY KEY,
  3      Tier_Name VARCHAR2(30) NOT NULL UNIQUE,
  4      Extra_Baggage_kg NUMBER(6,2) DEFAULT 0 NOT NULL,
  5      CONSTRAINT chk_tier_baggage CHECK (Extra_Baggage_kg >= 0)
  6  );

Table created.
```

```
SQL> CREATE TABLE CREW (
  2      Crew_ID NUMBER(10) PRIMARY KEY,
  3      Crew_Name VARCHAR2(60) NOT NULL,
  4      Role VARCHAR2(30) NOT NULL,
  5      Base_Location VARCHAR2(50) NOT NULL,
  6      CONSTRAINT chk_crew_role
  7          CHECK (Role IN ('PILOT', 'CO-PILOT', 'CABIN CREW', 'ENGINEER',
'GROUND STAFF'))
  8  );

Table created.
```

```
SQL> CREATE TABLE PASSENGER (
  2      Passenger_ID NUMBER(10) PRIMARY KEY,
  3      Passenger_Name VARCHAR2(60) NOT NULL,
  4      Passport_No VARCHAR2(20) NOT NULL UNIQUE,
  5      Tier_ID NUMBER(10) NOT NULL,
  6      Baggage_kg NUMBER(6,2) DEFAULT 0 NOT NULL,
  7      CONSTRAINT fk_passenger_tier
  8          FOREIGN KEY (Tier_ID) REFERENCES LOYALTY_TIER(Tier_ID),
  9      CONSTRAINT chk_passenger_baggage CHECK (Baggage_kg >= 0)
 10  );

Table created.
```

```
SQL> CREATE TABLE FLIGHT (
  2      Flight_ID NUMBER(10) PRIMARY KEY,
  3      Route_Code VARCHAR2(20) NOT NULL,
  4      Dep_Time TIMESTAMP NOT NULL,
  5      Arr_Time TIMESTAMP NOT NULL,
  6      Aircraft_ID NUMBER(10) NOT NULL,
  7      Flight_Status VARCHAR2(20) DEFAULT 'SCHEDULED' NOT NULL,
  8      CONSTRAINT fk_flight_aircraft
  9          FOREIGN KEY (Aircraft_ID) REFERENCES AIRCRAFT(Aircraft_ID),
 10      CONSTRAINT chk_flight_time CHECK (Arr_Time > Dep_Time),
 11      CONSTRAINT chk_flight_status
 12          CHECK (Flight_Status IN
 13          ('SCHEDULED', 'DEPARTED', 'ARRIVED', 'DELAYED', 'CANCELLED'))
 14  );

Table created.
```

```
SQL> CREATE TABLE FLIGHT_CREW (
  2      Assignment_ID NUMBER(10) PRIMARY KEY,
  3      Flight_ID NUMBER(10) NOT NULL,
  4      Crew_ID NUMBER(10) NOT NULL,
  5      Shift_Hours NUMBER(5,2) NOT NULL,
  6      CONSTRAINT fk_fc_flight
  7          FOREIGN KEY (Flight_ID) REFERENCES FLIGHT(Flight_ID),
  8      CONSTRAINT fk_fc_crew
  9          FOREIGN KEY (Crew_ID) REFERENCES CREW(Crew_ID),
 10      CONSTRAINT uq_flight_crew UNIQUE (Flight_ID, Crew_ID),
 11      CONSTRAINT chk_shift_hours CHECK (Shift_Hours > 0)
 12  );

Table created.
```

```
SQL> CREATE TABLE SPEC_CARGO (
  2      Cargo_ID NUMBER(10) PRIMARY KEY,
  3      Flight_ID NUMBER(10) NOT NULL,
  4      Cargo_Description VARCHAR2(100) NOT NULL,
  5      Weight_kg NUMBER(10,2) NOT NULL,
  6      Category VARCHAR2(20) NOT NULL,
  7      Req_Temp_C NUMBER(5,2),
  8      Current_Temp_C NUMBER(5,2),
  9      Tracking_Status VARCHAR2(20) DEFAULT 'IN TRANSIT' NOT NULL,
 10      CONSTRAINT fk_cargo_flight
 11          FOREIGN KEY (Flight_ID) REFERENCES FLIGHT(Flight_ID),
 12      CONSTRAINT chk_cargo_weight CHECK (Weight_kg > 0),
 13      CONSTRAINT chk_cargo_category
 14          CHECK (Category IN ('MEDICAL', 'PERISHABLE', 'HAZARDOUS', 'STANDARD')),
 15      CONSTRAINT chk_tracking_status
 16          CHECK (Tracking_Status IN ('RECEIVED', 'IN TRANSIT', 'DELIVERED', 'HOLD'))
 17  );

Table created.
```

```
SQL> CREATE TABLE EMISSIONS (
  2      Log_ID NUMBER(10) PRIMARY KEY,
  3      Flight_ID NUMBER(10) NOT NULL,
  4      Fuel_Liters NUMBER(12,2) NOT NULL,
  5      CO2_Tons NUMBER(10,3) NOT NULL,
  6      Logged_At TIMESTAMP DEFAULT CURRENT_TIMESTAMP NOT NULL,
  7      CONSTRAINT fk_emission_flight
  8          FOREIGN KEY (Flight_ID) REFERENCES FLIGHT(Flight_ID),
  9      CONSTRAINT chk_fuel CHECK (Fuel_Liters > 0),
 10      CONSTRAINT chk_co2 CHECK (CO2_Tons >= 0)
 11  );

Table created.
```

```
SQL> CREATE TABLE DISRUPTION (
  2      Event_ID NUMBER(10) PRIMARY KEY,
  3      Flight_ID NUMBER(10) NOT NULL,
  4      Delay_Min NUMBER(6) NOT NULL,
  5      Reason_Code VARCHAR2(20) NOT NULL,
  6      Event_Date TIMESTAMP NOT NULL,
  7      Event_Status VARCHAR2(20) DEFAULT 'OPEN' NOT NULL,
  8      CONSTRAINT fk_disruption_flight
  9          FOREIGN KEY (Flight_ID) REFERENCES FLIGHT(Flight_ID),
 10      CONSTRAINT chk_delay CHECK (Delay_Min >= 0),
 11      CONSTRAINT chk_reason_code
 12          CHECK (Reason_Code IN
 13          ('WEATHER', 'TECHNICAL', 'ATC', 'CREW', 'SECURITY', 'OTHER')),
 14      CONSTRAINT chk_event_status
 15          CHECK (Event_Status IN ('OPEN', 'RESOLVED'))
 16  );

Table created.
```

2. INSERTION - 5 ROWS PER TABLE

Aircraft table:

```
SQL> INSERT INTO AIRCRAFT
  2  (Aircraft_ID, Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status)
  3  VALUES (101, 'Airbus A320', 16800, 3.10, 'ACTIVE');

1 row created.

SQL> INSERT INTO AIRCRAFT
  2  (Aircraft_ID, Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status)
  3  VALUES (102, 'Boeing 737-800', 20800, 2.95, 'ACTIVE');

1 row created.

SQL>
SQL>
SQL> INSERT INTO AIRCRAFT
  2  (Aircraft_ID, Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status)
  3  VALUES (103, 'Airbus A321', 24000, 3.25, 'ACTIVE');

1 row created.

SQL> INSERT INTO AIRCRAFT
  2  (Aircraft_ID, Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status)
  3  VALUES (104, 'Boeing 787-9', 37000, 2.40, 'ACTIVE');

1 row created.
```

```
SQL> INSERT INTO AIRCRAFT
  2  (Aircraft_ID, Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status)
  3  VALUES (105, 'Airbus A330-300', 45000, 2.20, 'MAINTENANCE');

1 row created.
```

Loyalty_tier table:

```
SQL> INSERT INTO LOYALTY_TIER
  2  (Tier_ID, Tier_Name, Extra_Baggage_kg)
  3  VALUES (1, 'SILVER', 5);

1 row created.

SQL> INSERT INTO LOYALTY_TIER
  2  (Tier_ID, Tier_Name, Extra_Baggage_kg)
  3  VALUES (2, 'GOLD', 10);

1 row created.

SQL> INSERT INTO LOYALTY_TIER
  2  (Tier_ID, Tier_Name, Extra_Baggage_kg)
  3  VALUES (3, 'PLATINUM', 15);

1 row created.

SQL> INSERT INTO LOYALTY_TIER
  2  (Tier_ID, Tier_Name, Extra_Baggage_kg)
  3  VALUES (4, 'DIAMOND', 25);

1 row created.
```

```
SQL> INSERT INTO LOYALTY_TIER
  2  (Tier_ID, Tier_Name, Extra_Baggage_kg)
  3  VALUES (5, 'BASIC', 0);

1 row created.
```

Crew table:

```
SQL> INSERT INTO CREW
  2  (Crew_ID, Crew_Name, Role, Base_Location)
  3  VALUES (201, 'Ananya Sharma', 'PILOT', 'DELHI');

1 row created.

SQL>
SQL> INSERT INTO CREW
  2  (Crew_ID, Crew_Name, Role, Base_Location)
  3  VALUES (202, 'Rahul Mehta', 'CO-PILOT', 'MUMBAI');

1 row created.

SQL>
SQL> INSERT INTO CREW
  2  (Crew_ID, Crew_Name, Role, Base_Location)
  3  VALUES (203, 'Ishita Rao', 'CABIN CREW', 'BENGALURU');

1 row created.

SQL>
SQL> INSERT INTO CREW
  2  (Crew_ID, Crew_Name, Role, Base_Location)
  3  VALUES (204, 'Vikram Singh', 'ENGINEER', 'DELHI');

1 row created.

SQL>
SQL> INSERT INTO CREW
  2  (Crew_ID, Crew_Name, Role, Base_Location)
  3  VALUES (205, 'Neha Kapoor', 'CABIN CREW', 'HYDERABAD');

1 row created.
```

Flight table:

```
SQL> INSERT INTO FLIGHT
  2  (Flight_ID, Route_Code, Dep_Time, Arr_Time, Aircraft_ID, Flight_Status)
  3  VALUES (
  4  301, 'DEL-MUM',
  5  TIMESTAMP '2026-08-20 06:30:00',
  6  TIMESTAMP '2026-08-20 08:45:00',
  7  101, 'SCHEDULED');

1 row created.

SQL>
SQL> INSERT INTO FLIGHT
  2  (Flight_ID, Route_Code, Dep_Time, Arr_Time, Aircraft_ID, Flight_Status)
  3  VALUES (
  4  302, 'MUM-BLR',
  5  TIMESTAMP '2026-08-20 10:00:00',
  6  TIMESTAMP '2026-08-20 12:00:00',
  7  102, 'SCHEDULED');

1 row created.

SQL>
SQL> INSERT INTO FLIGHT
  2  (Flight_ID, Route_Code, Dep_Time, Arr_Time, Aircraft_ID, Flight_Status)
  3  VALUES (
  4  303, 'BLR-DEL',
  5  TIMESTAMP '2026-08-20 14:30:00',
  6  TIMESTAMP '2026-08-20 17:15:00',
  7  103, 'SCHEDULED');

1 row created.

SQL>
SQL> INSERT INTO FLIGHT
  2  (Flight_ID, Route_Code, Dep_Time, Arr_Time, Aircraft_ID, Flight_Status)
  3  VALUES (
  4  304, 'DEL-DXB',
  5  TIMESTAMP '2026-08-21 22:00:00',
  6  TIMESTAMP '2026-08-22 00:30:00',
  7  104, 'SCHEDULED');

1 row created.
```

```
SQL> INSERT INTO FLIGHT
  2  (Flight_ID, Route_Code, Dep_Time, Arr_Time, Aircraft_ID, Flight_Status)
  3  VALUES (
  4  305, 'MUM-SIN',
  5  TIMESTAMP '2026-08-21 09:00:00',
  6  TIMESTAMP '2026-08-21 17:00:00',
  7  105, 'DELAYED');

1 row created.
```

Flight_crew table:

```
SQL> INSERT INTO FLIGHT_CREW
  2  (Assignment_ID, Flight_ID, Crew_ID, Shift_Hours)
  3  VALUES (401, 301, 201, 4.00);

1 row created.

SQL>
SQL> INSERT INTO FLIGHT_CREW
  2  (Assignment_ID, Flight_ID, Crew_ID, Shift_Hours)
  3  VALUES (402, 301, 203, 4.00);

1 row created.

SQL>
SQL> INSERT INTO FLIGHT_CREW
  2  (Assignment_ID, Flight_ID, Crew_ID, Shift_Hours)
  3  VALUES (403, 302, 202, 3.50);

1 row created.

SQL>
SQL> INSERT INTO FLIGHT_CREW
  2  (Assignment_ID, Flight_ID, Crew_ID, Shift_Hours)
  3  VALUES (404, 303, 204, 5.00);

1 row created.

SQL>
SQL> INSERT INTO FLIGHT_CREW
  2  (Assignment_ID, Flight_ID, Crew_ID, Shift_Hours)
  3  VALUES (405, 304, 205, 6.00);

1 row created.
```

Passenger table:

```
SQL> INSERT INTO PASSENGER
  2  (Passenger_ID, Passenger_Name, Passport_No, Tier_ID, Baggage_kg)
  3  VALUES (501, 'Aarav Patel', 'P1234501', 2, 28);

1 row created.

SQL>
SQL> INSERT INTO PASSENGER
  2  (Passenger_ID, Passenger_Name, Passport_No, Tier_ID, Baggage_kg)
  3  VALUES (502, 'Meera Nair', 'P1234502', 3, 35);

1 row created.

SQL>
SQL> INSERT INTO PASSENGER
  2  (Passenger_ID, Passenger_Name, Passport_No, Tier_ID, Baggage_kg)
  3  VALUES (503, 'Rohan Verma', 'P1234503', 5, 20);

1 row created.

SQL>
SQL> INSERT INTO PASSENGER
  2  (Passenger_ID, Passenger_Name, Passport_No, Tier_ID, Baggage_kg)
  3  VALUES (504, 'Sara Khan', 'P1234504', 1, 25);

1 row created.

SQL>
SQL> INSERT INTO PASSENGER
  2  (Passenger_ID, Passenger_Name, Passport_No, Tier_ID, Baggage_kg)
  3  VALUES (505, 'Kabir Joshi', 'P1234505', 4, 42);

1 row created.
```

Spec_cargo table:

```
SQL> INSERT INTO SPEC_CARGO
  2  (Cargo_ID, Flight_ID, Cargo_Description, Weight_kg, Category,
  3  Req_Temp_C, Current_Temp_C, Tracking_Status)
  4  VALUES (601, 301, 'COVID Vaccine Shipment', 1200, 'MEDICAL',
  5  4, 4.20, 'IN TRANSIT');

1 row created.

SQL>
SQL> INSERT INTO SPEC_CARGO
  2  (Cargo_ID, Flight_ID, Cargo_Description, Weight_kg, Category,
  3  Req_Temp_C, Current_Temp_C, Tracking_Status)
  4  VALUES (602, 302, 'Fresh Seafood', 850, 'PERISHABLE',
  5  2, 2.40, 'IN TRANSIT');

1 row created.

SQL>
SQL> INSERT INTO SPEC_CARGO
  2  (Cargo_ID, Flight_ID, Cargo_Description, Weight_kg, Category,
  3  Req_Temp_C, Current_Temp_C, Tracking_Status)
  4  VALUES (603, 303, 'Industrial Chemicals', 1500, 'HAZARDOUS',
  5  NULL, NULL, 'RECEIVED');

1 row created.

SQL>
SQL> INSERT INTO SPEC_CARGO
  2  (Cargo_ID, Flight_ID, Cargo_Description, Weight_kg, Category,
  3  Req_Temp_C, Current_Temp_C, Tracking_Status)
  4  VALUES (604, 304, 'Fresh Produce', 2100, 'PERISHABLE',
  5  5, 5.10, 'IN TRANSIT');

1 row created.

SQL>
SQL> INSERT INTO SPEC_CARGO
  2  (Cargo_ID, Flight_ID, Cargo_Description, Weight_kg, Category,
  3  Req_Temp_C, Current_Temp_C, Tracking_Status)
  4  VALUES (605, 305, 'Electronic Equipment', 900, 'STANDARD',
  5  NULL, NULL, 'DELIVERED');

1 row created.
```

Emissions table:

```
SQL> INSERT INTO EMISSIONS
  2  (Log_ID, Flight_ID, Fuel_Liters, CO2_Tons, Logged_At)
  3  VALUES (701, 301, 4200, 13.020,
  4  TIMESTAMP '2026-08-20 09:00:00');

1 row created.

SQL>
SQL> INSERT INTO EMISSIONS
  2  (Log_ID, Flight_ID, Fuel_Liters, CO2_Tons, Logged_At)
  3  VALUES (702, 302, 3800, 11.210,
  4  TIMESTAMP '2026-08-20 12:30:00');

1 row created.

SQL>
SQL> INSERT INTO EMISSIONS
  2  (Log_ID, Flight_ID, Fuel_Liters, CO2_Tons, Logged_At)
  3  VALUES (703, 303, 4600, 14.950,
  4  TIMESTAMP '2026-08-20 17:45:00');

1 row created.

SQL>
SQL> INSERT INTO EMISSIONS
  2  (Log_ID, Flight_ID, Fuel_Liters, CO2_Tons, Logged_At)
  3  VALUES (704, 304, 12500, 30.000,
  4  TIMESTAMP '2026-08-22 01:00:00');

1 row created.

SQL>
SQL> INSERT INTO EMISSIONS
  2  (Log_ID, Flight_ID, Fuel_Liters, CO2_Tons, Logged_At)
  3  VALUES (705, 305, 14000, 31.500,
  4  TIMESTAMP '2026-08-21 18:00:00');

1 row created.
```

Disruption table:

```
SQL> INSERT INTO DISRUPTION
  2  (Event_ID, Flight_ID, Delay_Min, Reason_Code, Event_Date, Event_Status)
  3  VALUES (801, 301, 20, 'ATC',
  4  TIMESTAMP '2026-08-20 06:50:00', 'RESOLVED');

1 row created.

SQL>
SQL> INSERT INTO DISRUPTION
  2  (Event_ID, Flight_ID, Delay_Min, Reason_Code, Event_Date, Event_Status)
  3  VALUES (802, 302, 35, 'WEATHER',
  4  TIMESTAMP '2026-08-20 10:20:00', 'RESOLVED');

1 row created.

SQL>
SQL> INSERT INTO DISRUPTION
  2  (Event_ID, Flight_ID, Delay_Min, Reason_Code, Event_Date, Event_Status)
  3  VALUES (803, 303, 15, 'CREW',
  4  TIMESTAMP '2026-08-20 14:40:00', 'RESOLVED');

1 row created.

SQL>
SQL> INSERT INTO DISRUPTION
  2  (Event_ID, Flight_ID, Delay_Min, Reason_Code, Event_Date, Event_Status)
  3  VALUES (804, 304, 60, 'SECURITY',
  4  TIMESTAMP '2026-08-21 22:30:00', 'OPEN');

1 row created.

SQL>
SQL> INSERT INTO DISRUPTION
  2  (Event_ID, Flight_ID, Delay_Min, Reason_Code, Event_Date, Event_Status)
  3  VALUES (805, 305, 90, 'TECHNICAL',
  4  TIMESTAMP '2026-08-21 08:30:00', 'OPEN');

1 row created.
```

3.DELETION DEMONSTRATION:

Aircraft table:

```
SQL> INSERT INTO AIRCRAFT
  2  (Aircraft_ID, Model, Capacity_kg, Fuel_Efficiency, Aircraft_Status)
  3  VALUES (999, 'Test Aircraft', 10000, 2.50, 'ACTIVE');

1 row created.

SQL> DELETE FROM AIRCRAFT WHERE Aircraft_ID = 999;

1 row deleted.
```

Loyalty_tier table:

```
SQL> INSERT INTO LOYALTY_TIER
  2  (Tier_ID, Tier_Name, Extra_Baggage_kg)
  3  VALUES (999, 'TEST_TIER', 0);

1 row created.

SQL>
SQL> DELETE FROM LOYALTY_TIER WHERE Tier_ID = 999;

1 row deleted.
```

Crew table:

```
SQL> INSERT INTO CREW
  2  (Crew_ID, Crew_Name, Role, Base_Location)
  3  VALUES (999, 'Test Crew', 'CABIN CREW', 'DELHI');

1 row created.

SQL>
SQL> DELETE FROM CREW WHERE Crew_ID = 999;

1 row deleted.
```

Flight table:

```
SQL> INSERT INTO FLIGHT
  2  (Flight_ID, Route_Code, Dep_Time, Arr_Time, Aircraft_ID, Flight_Status)

  3  VALUES (999, 'TEST-TEST',
  4  TIMESTAMP '2026-08-25 08:00:00',
  5  TIMESTAMP '2026-08-25 10:00:00',
  6  101, 'SCHEDULED');

1 row created.

SQL>
SQL> DELETE FROM FLIGHT WHERE Flight_ID = 999;

1 row deleted.
```

Flight_crew table:

```
SQL> INSERT INTO FLIGHT_CREW
  2  (Assignment_ID, Flight_ID, Crew_ID, Shift_Hours)
  3  VALUES (999, 301, 205, 2.00);

1 row created.

SQL>
SQL> DELETE FROM FLIGHT_CREW WHERE Assignment_ID = 999;

1 row deleted.
```

Passenger table:

```
SQL> INSERT INTO PASSENGER
  2  (Passenger_ID, Passenger_Name, Passport_No, Tier_ID, Baggage_kg)
  3  VALUES (999, 'Test Passenger', 'P9999999', 5, 10);

1 row created.

SQL>
SQL> DELETE FROM PASSENGER WHERE Passenger_ID = 999;

1 row deleted.
```

Spec_cargo table:

```
SQL> INSERT INTO SPEC_CARGO
  2  (Cargo_ID, Flight_ID, Cargo_Description, Weight_kg, Category,
  3  Req_Temp_C, Current_Temp_C, Tracking_Status)
  4  VALUES (999, 301, 'Test Cargo', 100, 'STANDARD',
  5  NULL, NULL, 'RECEIVED');

1 row created.

SQL>
SQL> DELETE FROM SPEC_CARGO WHERE Cargo_ID = 999;

1 row deleted.
```

Emissions table:

```
SQL> INSERT INTO EMISSIONS
  2  (Log_ID, Flight_ID, Fuel_Liters, CO2_Tons, Logged_At)
  3  VALUES (999, 301, 1000, 3.00,
  4  TIMESTAMP '2026-08-20 11:00:00');

1 row created.

SQL>
SQL> DELETE FROM EMISSIONS WHERE Log_ID = 999;

1 row deleted.
```

Disruption table:

```
SQL> INSERT INTO DISRUPTION
  2  (Event_ID, Flight_ID, Delay_Min, Reason_Code, Event_Date, Event_Status)

  3  VALUES (999, 301, 10, 'OTHER',
  4  TIMESTAMP '2026-08-20 11:30:00', 'RESOLVED');

1 row created.

SQL>
SQL> DELETE FROM DISRUPTION WHERE Event_ID = 999;

1 row deleted.
```

4.UPDATION - ONE UPDATE ON EACH TABLE

Aircraft table:

```
SQL> UPDATE AIRCRAFT
  2  SET Fuel_Efficiency = 3.20
  3  WHERE Aircraft_ID = 101;

1 row updated.
```

Loyalty_tier table:

```
SQL> UPDATE LOYALTY_TIER
  2  SET Extra_Baggage_kg = 12
  3  WHERE Tier_ID = 2;

1 row updated.
```

Crew table:

```
SQL> UPDATE CREW
  2  SET Base_Location = 'MUMBAI'
  3  WHERE Crew_ID = 202;

1 row updated.
```

Flight table:

```
SQL> UPDATE FLIGHT
  2  SET Flight_Status = 'DELAYED'
  3  WHERE Flight_ID = 301;

1 row updated.
```

Flight_crew table:

```
SQL> UPDATE FLIGHT_CREW
  2  SET Shift_Hours = 4.50
  3  WHERE Assignment_ID = 401;

1 row updated.
```

Passenger table:

```
SQL> UPDATE PASSENGER
  2  SET Baggage_kg = 30
  3  WHERE Passenger_ID = 501;

1 row updated.
```

Spec_cargo table:

```
SQL> UPDATE SPEC_CARGO
  2  SET Current_Temp_C = 4.00
  3  WHERE Cargo_ID = 601;

1 row updated.
```

Emissions table:

```
SQL> UPDATE EMISSIONS
  2  SET CO2_Tons = 13.150
  3  WHERE Log_ID = 701;

1 row updated.
```

Disruption table:

```
SQL> UPDATE DISRUPTION
  2  SET Event_Status = 'RESOLVED'
  3  WHERE Event_ID = 804;

1 row updated.
```

5.SELECT QUERIES

Flight table:

```
SQL> select * from flight;

 FLIGHT_ID ROUTE_CODE          DEP_TIME                                                                    ARR_TIME
         AIRCRAFT_ID FLIGHT_STATUS
---------- ------------------- ------------------------------------------------------------------------- -------------------------------
-------------------------------------- ---------- --------------------
       301 DEL-MUM             20-AUG-26 06.30.00.000000 AM                                              20-AUG-26 08.45.00.000000 AM
           101 DELAYED
       302 MUM-BLR             20-AUG-26 10.00.00.000000 AM                                              20-AUG-26 12.00.00.000000 PM
           102 SCHEDULED
       303 BLR-DEL             20-AUG-26 02.30.00.000000 PM                                              20-AUG-26 05.15.00.000000 PM
           103 SCHEDULED
       304 DEL-DXB             21-AUG-26 10.00.00.000000 PM                                              22-AUG-26 12.30.00.000000 AM
           104 SCHEDULED
       305 MUM-SIN             21-AUG-26 09.00.00.000000 AM                                              21-AUG-26 05.00.00.000000 PM
           105 DELAYED
```

```
SQL> SELECT Flight_ID, Route_Code, Flight_Status
  2  FROM FLIGHT
  3  WHERE Flight_Status = 'DELAYED';

 FLIGHT_ID ROUTE_CODE           FLIGHT_STATUS
---------- -------------------- --------------------
       301 DEL-MUM              DELAYED
       305 MUM-SIN              DELAYED
```

Passenger table:

```
SQL> SELECT Passenger_ID, Passenger_Name, Passport_No, Baggage_kg
  2  FROM PASSENGER;

PASSENGER_ID PASSENGER_NAME
------------ ------------------------------------------------------------
PASSPORT_NO          BAGGAGE_KG
-------------------- ----------
         501 Aarav Patel
P1234501                     30

         502 Meera Nair
P1234502                     35

         503 Rohan Verma
P1234503                     20


PASSENGER_ID PASSENGER_NAME
------------ ------------------------------------------------------------
PASSPORT_NO          BAGGAGE_KG
-------------------- ----------
         504 Sara Khan
P1234504                     25

         505 Kabir Joshi
P1234505                     42
```

6.SINGLE-ROW FUNCTIONS

Aircraft table:

```
SQL> SELECT Aircraft_ID,
  2         Model,
  3         LENGTH(Model) AS Model_Length
  4  FROM AIRCRAFT;

AIRCRAFT_ID MODEL                                              MODEL_LENGTH
----------- -------------------------------------------------- ------------
        101 Airbus A320                                                11
        102 Boeing 737-800                                             14
        103 Airbus A321                                                11
        104 Boeing 787-9                                               12
        105 Airbus A330-300                                            15
```

Crew table:

```
SQL> SELECT Crew_ID,
  2         LOWER(Crew_Name) AS Crew_Name_Lower
  3  FROM CREW;

   CREW_ID CREW_NAME_LOWER
---------- --------------------------------------------------
       201 ananya sharma
       202 rahul mehta
       203 ishita rao
       204 vikram singh
       205 neha kapoor
```

Passenger table:

```
SQL> SELECT Passenger_ID,
  2         UPPER(Passenger_Name) AS Passenger_Name_Upper
  3  FROM PASSENGER;

PASSENGER_ID PASSENGER_NAME_UPPER
------------ --------------------------------------------------
         501 AARAV PATEL
         502 MEERA NAIR
         503 ROHAN VERMA
         504 SARA KHAN
         505 KABIR JOSHI
```

```
SQL> SELECT Passenger_ID,
  2         Passenger_Name || ' - ' || Passport_No AS Passenger_Details
  3  FROM PASSENGER;

PASSENGER_ID PASSENGER_DETAILS
------------ --------------------------------------------------------------------------------
         501 Aarav Patel - P1234501
         502 Meera Nair - P1234502
         503 Rohan Verma - P1234503
         504 Sara Khan - P1234504
         505 Kabir Joshi - P1234505
```

Spec_cargo table:

```
SQL> SELECT Cargo_ID,
  2         Cargo_Description,
  3         COALESCE(Current_Temp_C, 0) AS Display_Temperature
  4  FROM SPEC_CARGO;

  CARGO_ID CARGO_DESCRIPTION                                       DISPLAY_TEMPERATURE
---------- --------------------------------------------------------------------------------
       601 COVID Vaccine Shipment                                                    4
       602 Fresh Seafood                                                           2.4
       603 Industrial Chemicals                                                      0
       604 Fresh Produce                                                           5.1
       605 Electronic Equipment                                                      0
```

Emissions table:

```
SQL> SELECT Flight_ID,
  2         ROUND(CO2_Tons, 2) AS Rounded_CO2_Tons
  3  FROM EMISSIONS;

 FLIGHT_ID ROUNDED_CO2_TONS
---------- ----------------
       301            13.15
       302            11.21
       303            14.95
       304               30
       305             31.5
```

7.AGGREGATE FUNCTIONS

Flight table:

```
SQL> SELECT COUNT(*) AS Total_Flights
  2  FROM FLIGHT;

TOTAL_FLIGHTS
-------------
            5
```

Spec_cargo table:

```
SQL> SELECT SUM(Weight_kg) AS Total_Cargo_Weight
  2  FROM SPEC_CARGO;

TOTAL_CARGO_WEIGHT
------------------
              6550
```

Emissions table:

```
SQL> SELECT MAX(CO2_Tons) AS Maximum_CO2_Tons
  2  FROM EMISSIONS;

MAXIMUM_CO2_TONS
----------------
            31.5
SQL> SELECT MIN(Fuel_Liters) AS Minimum_Fuel_Used
  2  FROM EMISSIONS;

MINIMUM_FUEL_USED
-----------------
             3800
```

Disruption table:

```
SQL> SELECT ROUND(AVG(Delay_Min), 2) AS Average_Delay_Minutes
  2  FROM DISRUPTION;

AVERAGE_DELAY_MINUTES
---------------------
                   44
```

8.GROUP BY:

Flight table:

```
SQL> SELECT Flight_Status,
  2         COUNT(*) AS Flight_Count
  3  FROM FLIGHT
  4  GROUP BY Flight_Status;

FLIGHT_STATUS         FLIGHT_COUNT
--------------------- ------------
DELAYED                          2
SCHEDULED                        3
```

Spec_cargo table:

```
SQL> SELECT Category,
  2         SUM(Weight_kg) AS Total_Weight_kg
  3  FROM SPEC_CARGO
  4  GROUP BY Category;

CATEGORY             TOTAL_WEIGHT_KG
-------------------- ---------------
MEDICAL                         1200
PERISHABLE                      2950
HAZARDOUS                       1500
STANDARD                         900
```

Disruption table:

```
SQL> SELECT Reason_Code,
  2         SUM(Delay_Min) AS Total_Delay_Minutes
  3  FROM DISRUPTION
  4  GROUP BY Reason_Code;

REASON_CODE          TOTAL_DELAY_MINUTES
-------------------- -------------------
ATC                                   20
WEATHER                               35
CREW                                  15
SECURITY                              60
TECHNICAL                             90
```

9.HAVING:

Spec_cargo table:

```
SQL> SELECT Category,
  2         SUM(Weight_kg) AS Total_Weight_kg
  3  FROM SPEC_CARGO
  4  GROUP BY Category
  5  HAVING SUM(Weight_kg) > 1000;

CATEGORY             TOTAL_WEIGHT_KG
-------------------- ---------------
MEDICAL                         1200
PERISHABLE                      2950
HAZARDOUS                       1500
```

Disruption table:

```
SQL> SELECT Reason_Code,
  2         SUM(Delay_Min) AS Total_Delay_Minutes
  3  FROM DISRUPTION
  4  GROUP BY Reason_Code
  5  HAVING SUM(Delay_Min) > 30;

REASON_CODE          TOTAL_DELAY_MINUTES
-------------------- -------------------
WEATHER                               35
SECURITY                              60
TECHNICAL                             90
```

10.SUBQUERIES:

Aircraft table:

```
SQL> SELECT Aircraft_ID, Model, Fuel_Efficiency
  2  FROM AIRCRAFT
  3  WHERE Fuel_Efficiency > (
  4      SELECT AVG(Fuel_Efficiency)
  5      FROM AIRCRAFT
  6  );

AIRCRAFT_ID MODEL                                          FUEL_EFFICIENCY
----------- ---------------------------------------------- ---------------
        101 Airbus A320                                              3.2
        102 Boeing 737-800                                          2.95
        103 Airbus A321                                             3.25
```

Passenger table:

```
SQL> SELECT Passenger_ID, Passenger_Name, Baggage_kg
  2  FROM PASSENGER
  3  WHERE Baggage_kg > (
  4      SELECT AVG(Baggage_kg)
  5      FROM PASSENGER
  6  );

PASSENGER_ID PASSENGER_NAME                                                BAGGAGE_KG
------------ ------------------------------------------------------------ ----------
         502 Meera Nair                                                             35
         505 Kabir Joshi                                                            42
```

Emissions table:

```
SQL> SELECT Flight_ID, CO2_Tons
  2  FROM EMISSIONS
  3  WHERE CO2_Tons > (
  4      SELECT AVG(CO2_Tons)
  5      FROM EMISSIONS
  6  );

 FLIGHT_ID   CO2_TONS
---------- ----------
       304         30
       305       31.5
```

11.JOIN QUERIES

Aircraft table AND Flight table:

```
SQL> SELECT F.Flight_ID,
  2         F.Route_Code,
  3         A.Model,
  4         A.Capacity_kg
  5  FROM FLIGHT F
  6  JOIN AIRCRAFT A
  7  ON F.Aircraft_ID = A.Aircraft_ID;

 FLIGHT_ID ROUTE_CODE          MODEL                                              CAPACITY_KG
---------- ------------------- -------------------------------------------------- -----------
       301 DEL-MUM             Airbus A320                                              16800
       302 MUM-BLR             Boeing 737-800                                           20800
       303 BLR-DEL             Airbus A321                                              24000
       304 DEL-DXB             Boeing 787-9                                             37000
       305 MUM-SIN             Airbus A330-300                                          45000
```

Loyalty_tier table AND Passenger:

```
SQL> SELECT P.Passenger_Name,
  2         P.Baggage_kg,
  3         L.Tier_Name,
  4         L.Extra_Baggage_kg
  5  FROM PASSENGER P
  6  JOIN LOYALTY_TIER L
  7  ON P.Tier_ID = L.Tier_ID;

PASSENGER_NAME                                             BAGGAGE_KG TIER_NAME                     EXTRA_BAGGAGE_KG
---------------------------------------------------------- ---------- ------------------------------ ----------------
Sara Khan                                                          25 SILVER                                      5
Aarav Patel                                                        30 GOLD                                       12
Meera Nair                                                         35 PLATINUM                                   15
Kabir Joshi                                                        42 DIAMOND                                    25
Rohan Verma                                                        20 BASIC                                       0
```

Crew table AND Flight table AND Flight_crew table:

```
SQL> SELECT F.Flight_ID,
  2         F.Route_Code,
  3         C.Crew_Name,
  4         C.Role,
  5         FC.Shift_Hours
  6  FROM FLIGHT F
  7  JOIN FLIGHT_CREW FC
  8  ON F.Flight_ID = FC.Flight_ID
  9  JOIN CREW C
 10  ON FC.Crew_ID = C.Crew_ID;

 FLIGHT_ID ROUTE_CODE          CREW_NAME                                          ROLE                           SHIFT_HOURS
---------- ------------------- -------------------------------------------------- ------------------------------ -----------
       301 DEL-MUM             Ananya Sharma                                      PILOT                                  4.5
       302 MUM-BLR             Rahul Mehta                                        CO-PILOT                               3.5
       301 DEL-MUM             Ishita Rao                                         CABIN CREW                               4
       303 BLR-DEL             Vikram Singh                                       ENGINEER                                 5
       304 DEL-DXB             Neha Kapoor                                        CABIN CREW                               6
```

Spec_cargo table AND Flight table:

```
SQL> SELECT F.Flight_ID,
  2         F.Route_Code,
  3         S.Cargo_Description,
  4         S.Category,
  5         S.Weight_kg,
  6         S.Current_Temp_C,
  7         S.Tracking_Status
  8  FROM FLIGHT F
  9  JOIN SPEC_CARGO S
 10  ON F.Flight_ID = S.Flight_ID;

 FLIGHT_ID ROUTE_CODE          CARGO_DESCRIPTION                                                                                  CATEGORY
  WEIGHT_KG CURRENT_TEMP_C TRACKING_STATUS
---------- ---------- ---------- --------------------------------------------------------------------------------------------------- ----------
---------- ----------- ---------- --------------------
       301 DEL-MUM               COVID Vaccine Shipment                                                                             MEDICAL   1
200             4 IN TRANSIT
       302 MUM-BLR               Fresh Seafood                                                                                      PERISHABLE
 850           2.4 IN TRANSIT
       303 BLR-DEL               Industrial Chemicals                                                                               HAZARDOUS 1
500             RECEIVED
       304 DEL-DXB               Fresh Produce                                                                                      PERISHABLE
2100            5.1 IN TRANSIT
       305 MUM-SIN               Electronic Equipment                                                                               STANDARD
900             DELIVERED
```

Flight table AND Aircraft table:

```
SQL> SELECT F.Flight_ID,
  2         F.Route_Code,
  3         A.Model,
  4         E.Fuel_Liters,
  5         E.CO2_Tons
  6  FROM FLIGHT F
  7  JOIN AIRCRAFT A
  8  ON F.Aircraft_ID = A.Aircraft_ID
  9  JOIN EMISSIONS E
 10  ON F.Flight_ID = E.Flight_ID;

 FLIGHT_ID ROUTE_CODE          MODEL                                              FUEL_LITERS   CO2_TONS
---------- ------------------- -------------------------------------------------- ----------- ----------
       301 DEL-MUM             Airbus A320                                              4200      13.15
       302 MUM-BLR             Boeing 737-800                                           3800      11.21
       303 BLR-DEL             Airbus A321                                              4600      14.95
       304 DEL-DXB             Boeing 787-9                                            12500         30
       305 MUM-SIN             Airbus A330-300                                         14000       31.5
```

Disruption table AND Flight table:

```
SQL> SELECT F.Route_Code,
  2         D.Delay_Min,
  3         D.Reason_Code,
  4         D.Event_Status
  5  FROM FLIGHT F
  6  JOIN DISRUPTION D
  7  ON F.Flight_ID = D.Flight_ID
  8  ORDER BY D.Delay_Min DESC;

ROUTE_CODE          DELAY_MIN REASON_CODE         EVENT_STATUS
------------------- ---------- ------------------- --------------------
MUM-SIN                     90 TECHNICAL           OPEN
DEL-DXB                     60 SECURITY            RESOLVED
MUM-BLR                     35 WEATHER             RESOLVED
DEL-MUM                     20 ATC                 RESOLVED
BLR-DEL                     15 CREW                RESOLVED
```